

- Q.27 Define wet bulb and dry bulb temperature.
- Q.28 Draw Humidity Chart and mark to show different parameters of Air Water mixture.
- Q.29 Differentiate between humidification and dehumidification with example
- Q.30 Draw and explain any two cooling tower arrangements.
- Q.31 Write the use of tower packing with their types and properties.
- Q.32 Discuss the working of spray chambers in brief.
- Q.33 Draw and explain in brief rate of drying curve.
- Q.34 Explain the working of rotary dryer with neat diagram.
- Q.35 Explain problem Flooding Weeping encountered during operation of column.

#### SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Define Drying. Explain construction and working of Tray dryer with its application.
- Q.37 Write about different types of tower packing. Explain the different problems encountered in packed towers.
- Q.38 With the help of neat diagram explain the working principle of natural draft cooling tower

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#### 4th Sem / Chemical Engineering, P&P Sub : Mass Transfer-I

Time : 3Hrs.

M.M. : 100

#### SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Diffusion is movement of molecules due to
- Pressure gradient
  - Temperature gradient
  - Concentration gradient
  - Velocity gradient
- Q.2 Eddy diffusion occurs mainly in
- Solids
  - Liquids
  - Gases
  - Turbulent flow
- Q.3 Which law is used for gas absorption equilibrium
- Fick's law
  - Henry's law
  - Fourier's law
  - Newton's law
- Q.4 Humidity is related to
- Heat transfer
  - Moisture in air
  - Pressure
  - Density

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Q.5 Wet bulb temperature is measured by

- a) Ordinary thermometer
- b) Hygrometer
- c) Wet thermometer
- d) Barometer

Q.6 NTU means

- a) Number of Transfer Units
- b) Number of Thermal Units
- c) Normal Transfer Unit
- d) None

Q.7 Drying is removal of

- a) Air
- b) Gas
- c) Moisture
- d) Heat

Q.8 Weeping in packed column is due to

- a) High vapor flow
- b) Low vapor flow
- c) No liquid flow
- d) High temperature

Q.9 Humidification involved

- a) Removal of moisture
- b) Addition of moisture
- c) Cooling of Air
- d) Compression of Air

Q.10 Critical moisture content is related to

- a) Adsorption
- b) Drying
- c) Absorption
- d) Diffusion

## SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define diffusion coefficient.

Q.12 Write one application of absorption.

Q.13 What is dew point?

Q.14 Write full form of HETP.

Q.15 Define humid heat.

Q.16 Name one cooling tower arrangement.

Q.17 What is desorption?

Q.18 Define free moisture.

Q.19 Write one use of humidity chart.

Q.20 Define drying.

## SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Write a short note on diffusion in mass transfer.

Q.22 Write in brief about any two mass transfer operations.

Q.23 Define diffusivity and also state Fick's law of diffusion.

Q.24 Discuss Important factors for choosing a solvent for absorption.

Q.25 Write Raoult's law and Henry's law.

Q.26 Discuss in brief about two film theory of mass transfer.